Atoms and Molecules:

- Around 500 B.C, Indian Philosopher named Maharishi Kanad postulated that matter can be divided into extremely small particles, which can't be divided further and are called as Parmanus.

- Greek Philosphers - Democritus and Leucippus gave same idea but called these indivisible particles as "Atoms".

Atom - "That can't be cut".

→ **Laws of Chemical Combination:-**

1) **Law of Conservation of mass:-** (By Antonie Lavoiser)

- During any physical or chemical change the total mass of reactant is equal to total mass of products.

'OR'

"Mass can neither be created nor destroyed".

e.g. Mercuric oxide $\xrightarrow{heat}$ mercury + oxygen
100g → 92.6g + 7.4g

- Total mass of Reactants = Total Mass of Products
100g = 92.6 + 7.4 = 100g

Q) If 10g of calcium carbonate ($CaCO_3$) on heating gave 4.4g of carbon dioxide (CO_2) and 5.6g of calcium oxide (CaO). Show these observations are in agreement with law of conservation of mass?

Sol: $CaCO_3 \xrightarrow{Heat\ (\Delta)} CO_2 + CaO$
10g → 4.4g + 5.6g

- Acc. to Law of conservation of mass of reactants = Total mass of products.

$$10g = 4.4 + 5.6$$

- Yes it follows Law of Conservation of mass.

- <u>Law of Constant Composition 'OR' Definite Proportion</u>

☆- Proposed by Joseph Proust.

- A pure chemical compound is always made up of same elements that are combined in same proportion by mass.

e.g. Water from any source - River, Lake, Ocean, Rain, etc. is always made up of hydrogen and Oxygen in ratio H_2O

$$2:16 = 1:8.$$

Illy. CO_2 i) $CaCO_3 \xrightarrow{\Delta} CO_2 + CaO$

ii) $C + O_2 \longrightarrow CO_2$

In case (i) and (ii) CO_2 prepared by different method but elements ~~are~~ and ratio is same.

$$CO_2 \rightarrow 12:32$$
$$\Rightarrow 3:8$$

Q) When 1.375g of Cupric Oxide (CuO) was reduced on heating in a current of hydrogen the weight of copper (Cu) that remained was 1.098g. In another experiment 1.179g of copper was dissolved in nitric acid and resulting copper nitrate converted into cupric oxide by ignition. The weight of cupric oxide formed was 1.476g. Show that these results illustrate the law of ~~constant~~ constant proportion.

Sol In experiment no. 1

Mass of $CuO = 1.375g$
Mass of $Cu = 1.098g$
Mass of oxygen $= 1.375g - 1.098g = 0.277g$

$\% \text{ of } Cu = \frac{\text{Mass of } Cu}{\text{Mass of } CuO} \times 100 = \frac{1.098 \times 100}{1.375}$
$= 79.85\%$
$= 80\%$

$\% \text{ of Oxygen} = \frac{\text{Mass of oxygen}}{\text{Mass of } CuO} \times 100 = \frac{0.277 \times 100}{1.375}$
$= 20\%$

In experiment no. 2

Mass of $CuO = 1.476$
Mass of $Cu = 1.179$
Mass of oxygen $= 1.476 - 1.179 = 0.297$

$\% \text{ of } Cu = \frac{\text{Mass of } Cu}{\text{Mass of } CuO} = \frac{1.179}{1.476} \times 100$
$= 79.8\%$
$= 80\%$

% of oxygen $= \frac{\text{Mass of oxygen}}{\text{Mass of CuO}} \times 100$

$= \frac{0.297}{1.476} \times 100$

$= 20\%$

∴ Law of constant composition followed.

⇒ <u>Dalton's Atomic Theory</u>:-

• <u>Proposed by John Dalton</u>.

- John Dalton studied structure of matter and proposed following points:-

1) Matter is made up of extremely small particles called as Atoms.

2) Atoms of an element are similar in every respect, i.e; Mass, shape, size, etc.

3) Atoms of different elements are different in every respect i.e; mass, size, shape etc.

4) Atome is the smallest particle that takes part in a chemical reaction.

5) Atoms combine together in simple whole no. ratio to form molecules.

6) Atoms are indivisible i.e; they can't be divided further.

⇒ <u>Limitations of Dalton's Atomic theory:</u>

1) This theory did not explain why the atoms of different elements are different i.e; this theory did not talk about internal structure of Atom.

2) Why and how atoms combine and form molecules

3) Atom is indivisible according to this theory but Atom can be divided.

⇒ **Atom:-** It is the smallest particle that may or may not have independent existence.

e.g. Atoms that exist independently = Na, C, K, B, etc.

Atoms that exist in combined form = O_2, H_2, N_2, etc.

⇒ **Symbols of Atoms of different elements:-**

- ~~Joh~~ John Dalton proposed some nortification to represent symbols of Atoms.

Atoms	symbols.
e.g.:- Carbon →	●
Oxygen →	○
Hydrogen →	⊙

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- These symbols were later discarded.

⇒ Berzelius proposed that the symbols should include letters from name of atom.

- IUPAS - International Union of pure and applied chemistry; developed following symbols:

A) Elements that use first letter of name as symbol:

i) Hydrogen - H
ii) Oxygen - O
iii) Nitrogen - N
iv) Carbon - C, etc.

B) First two letters in symbols:

Aluminium - Al
Helium - He
Lithium - Li
Silicon - Si
Neon - Ne
Argon - Ar, etc.

c) First and third letter of name:-

Magnesium - Mg
Chlorine - Cl
zinc - zn , etc.

d) Symbols from Latin Names:-

Iron - ~~Ferr~~ Ferrum - Fe
Lead - Plumbum - Pb
Gold - Aurum - Au
Sodium - Natrium - Na
Potassium - kalium - K
Tin - stannum - Sn

★ ⇒ Atomic Mass:-

- In Ancient times isolation of ~~som~~ single atom and calculation of its mass was difficult.
- Mass of any atom was calculated with respect to some standard or refrence value, so the

mass calculated is called as Relative Atomic mass.

→ Relative Atomic mass calculated with respect to some standard value.

- Scale on which these masses were expressed is called as Atomic Mass scale.

→ The refrence chosen for calculation of relative atomic mass.

1. Hydrogen
But mass of some atoms came out to be fractional so it was discarded.

2. Oxygen
But mass of some atoms showed variation using oxygen as standard so it was also discarded.

3. Finally IUPAS set carbon as standard and

mass of other atoms was calculated using carbon as 12 units on atomic mass scale.

⇒ **Atomic Mass:** It is defined as relative atomic mass of an atom with respect to C-12 taken as 12 units.

⇒ **Unit:-** amu → Atomic mass unit

e.g. For Hydrogen - 1 amu. Oxygen = 8 amu.

⇒ **Atomic Mass Unit - (amu):-**

- Atomic mass unit is defined as $\frac{1}{12^{th}}$ mass of carbon atom.

$$1 amu = 1.66 \times 10^{-24} g$$

⇒ **Molecules:-**

- Molecules is defined as the smallest particle of matter that exists freely, the properties of that substance. Individual properties of atom are not present in molecule.

e.g CO_2, H_2O, NH_3, etc.

Molecules ~~of~~ - 2 Types

Molecule of Element. | Molecule of compound

1) Molecule of Element:-

- Also called as ~~Homot~~ HOMOATOMIC Molecules.
- These are the molecules that are made up of same type of atoms.

e.g O_2, O_3, P_4 etc.

2) Molecule of Compound:-

- Also called as HETEROATOMIC molecules.
- These are the molecules that are made up of

different type of atoms.

e.g CO_2, H_2O, NH_3, etc.

⇒ **Atomicity**:- It is the no. of atoms present in a molecule.

Diatomic → 2 atoms
Triatomic → 3 atoms
Tetraatomic → 4 atoms
Pentaatomic → 5 atoms.

(e.g) Homoatomic	Heteroatomic
• Monoatomic - He, Ne, Ar, Xe, Noble gases.	• Diatomic: NaCl, KCl, CO
	• Triatomic: H_2O, CO_2
• Diatomic:- O_2, H_2, N_2	• Tetraatomic:- NH_3
• Triatomic : O_3	
• Tetraatomic : P_4	

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⇒ **Ions:** Ion is defined as an atom or group of atoms carrying positive or negative charge.

e.g Na^+, Cl^-, NH_4^+, OH^-, etc.

- Ions are of two types, 1) ~~Atom~~ Cation
  2) Anion.

⇒ **Cation:** Ion carrying positive charge.

e.g Na^+, Ca^{+2}, Al^{+3}, NH_4^+, etc.

⇒ **Anion:** Ion carrying negative charge.

e.g Cl^-, OH^-, O^{-2}, etc.

→ Cation is formed when an atom looses electron.

→ Atom is formed when a atom gains ~~electron~~ electron.

⇒ Ions can be Monoatomic or Polyatomic.

⇒ **Monoatomic Ion:** Ion containing ~~more~~ only one atom. e.g Na^+, Cl^-, O^{-2}, etc.

⇒ **Polyatomic Ion**:- Ion containing more than one atom. e.g CO_3^{-2}, NH_4^+, OH^-, etc.

⇒ **Molecular Mass**:-

- Molecular mass may be defined as relative mass of a molecule with respect to mass of carbon-12 taken as 12 units.

⇒ **Calculation**:- Molecular Mass = Sum of R.A.M Present in Molecule.

e.g H_2.

Molecular Mass = 2 x atomic mass of H.

= 2 x 1

= 2 units.

CO_2.

Molecar mass = C + O + O

= 12 + 16 + 16

= 44u.

Q) Calculate molecular Mass of the following.

1) $CaCO_3$ (2) $C_6H_{12}O_6$
↓ ↓
Calcium Carbonate Glucose.

1) Molecular Mass of Calcium Carbonate

= Atomic mass (Ca) + Atomic mass (C) + 3 × A.M

$= 40u + 12u + 3 \times 16u$

$= 100u$

2) $C_6H_{12}O_6$:- $6 \times C + 12 \times H + 6 \times O$

$= 6 \times 12 + 12 \times 1 + 6 \times 16$

$= 180u.$

⇒ Valency:-

- Valency is defined as combining tendency of an element.

- In case of ions Valency is defined as units of positive or negative charge present on ion.

⇒ Valency of some elements:-

Element	Valency
Hydrogen	1
Sodium	1
Chlorine	1
Oxygen	2
Nitrogen	3
Carbon	4

⇒

Ions	Valency
H^+	1
Na^+	1
Ca^{+2}	2
O^{-2}	2
Al^{+3}	3
(Hydrogen Ion) OH^-	1
Cl^-	1
(Carbonate) CO_3^{-2}	2
(Sulphate) SO_4^{-2}	2

→ Chemical Formula:-

- It is defined as brief representation of a chemical compound in terms of symbols of constituents elements.

e.g.

Chemical Compound	Chemical Formulla
1) Water	H_2O
2) Carbon dioxide	CO_2.
3) Ammonia	NH_3
4) Calcium Carbonate	$CaCO_3$.

⇒ Writing of Chemical Formula-

- Write Symbol and Valency or charge of ion in manner, and apply criss cross.

H O
1 2

- Then the valency ~~will~~ will appear in subscript of the symbol.

→ If there is any common factor between Valencies then the Valencies are divisible by that ~~con comon~~ common factor.

⇒ <u>Important Chemical Formula</u>

1- Water.
→ Symbol :- H O
→ Valency:- 1 2
→ Formula:- H_2O

2) Carbon dioxide

- Symbol:- C O
- Valency:- $\cancel{4}\,2$ $\cancel{2}\,1$
- Formula:- CO_2

3) Methane

- Symbol :- C H
- Velency 4 1

- Formula - CH_4

4) Amonia.

- Symbol: N H
- Velency: 3 1
- Formula. NH_3

5) Carbon Monoxide

- Symbol. C O
- Valency - $\cancel{4}_2$ $\cancel{2}_1$
- Formula CO_2

6) Sodium Bromide

- Symbol - Na^+ Br^-
- Valency - 1 1
- Formula - NaBr

7) Magnesium Sulfhate

- Symbol - Mg^{+2} SO_4^{-2}
- Valency. $\cancel{2}$ $\cancel{2}$
- Formula. $MgSO_4$

8) Sulphuric Acid

- Symbol. H^{+} SO_4^{-2}
- Valency. 1 2
- Formula- H_2SO_4

9) Ammonium Sulphate

- Symbol- NH_4^{+} SO_4^{-2}
- Valency- 1 2
- Formula $(NH_4)_2CO_4$

⇒ Gram Atomic Mass (GAM) and Gram Molecular mass (GAM):-

1) Gram Atomic Mass (GAM)-

- Atomic mass expressed in grams.

e.g. H = 1u   C = 12u   N = 14u
GAM = 1g   GAM = 12g   GAM = 14g.

2) Gram Molecular Mass:- Molecular mass expressed

in grams.

e.g $CO_2 = 44u$ $H_2O = 18u$
~~GAG~~ GMM = 44g GMM = 18g.

⇒ **Mole Concept-**

- Since it was not possible to count the no. of atoms present in a sample.
- So this count was related to mass of sample.

→ It was discovered that no. of atoms present in GAM of C i.e; $12g = 6.022 \times 10^{23}$ atoms of carbon.

→ Similarly GAM of oxygen = 16g contained:- 6.022×10^{23} atoms of O.

→ This no. 6.022×10^{23} is called as Avogadro's no.

(N_a or N) after name of Ameder Avogadro.

→ Avogadro's no. is expressed in terms of Mole.

*⇒ 1 mole = 6.022×10^{23} particles.

- 1 mole of hydrogen atoms = 6.022×10^{23} atoms of hydrogen.

- 1 mole of water molecules = 6.022×10^{23} molecules of Water

- 1 mole of Sodium ion = 6.022×10^{23} Sodium ion.

⇒ <u>Mole and Mass</u> -

- Mole of ~~ato~~ atoms is defined as that amount of substance which has mass equal to Gram atomic mass or Gram molecular mass.

- 1 mole of oxygen molecule (O_2):

→ GMM = 32g = 6.022×10^{23} molecules.

- 1 mole of hydrogen atoms

$\rightarrow GAM = 1g = 6.022 \times 10^{23}$ atoms.

⇒ **Mole and no. of particles :-**

- One mole in terms of no. of particles is

1 mole = 6.022×10^{23} particles.
Particles ⇒ atoms/Molecules/ions.

⇒ **Mole ~~of~~ and Volume.**

- For gaseous substances:-

1 mole of any gas at STP (Standard temperature and pressure) has volume equal to 22.4 litres

⇒ [STP:- Temperature = 0°c. Pressure = 1 atmo.].

⇒ **Important Formulas**

1). No. of moles = $\dfrac{\text{Given mass in grams}}{\text{Molecular Mass}}$

2) No. of particles in given mass

$$\text{No. of particles} = \frac{\text{Given mass} \times \text{Avagadro's no.}}{\text{Molecular Mass}}$$

→ Numericals

Q: Calculate the no. of moles in i) 100g of iron (Fe) (Atomic mass Fe = 56u). ii) 100g of Water (Molecular mass = 18u)

Sol) (i) $\text{No. of moles} = \frac{\text{Given mass of Fe}}{\text{Atomic mass}}$

$\text{No. of moles} = \frac{100g}{56g} = 1.78$ moles.

(ii) $\text{No. of moles} = \frac{\text{Given mass of water}}{\text{Molecular Mass}}$

$= \frac{100g}{18} = 5.55$ moles.

Q:- If one mole of Carbon weighs 12gram. What is the mass of ~~the~~ 1atom of carbon in grams?

sol:- 1 mole = 6.022×10^{23} atoms = 12g.

6.022×10^{23} atoms weighs = 12g

∴ 1 atom weigh = $\frac{12}{6.022 \times 10^{23}} \times 1$ = 1.993×10^{-23} g.

Q: Calculate the mass of 0.5 moles of silver (Ag).

Atomic mass of Ag = 108u.

Sol:- GAM of Ag = 108g

∴ 1 mole of Ag = 108g

∴ 0.5 mole of Ag = 108×0.5

= 54g.

Q: Calculate the no. of Atoms ~~of gold~~ in 1g of gold. Atomic mass of Gold (Au) = 197u.

Sol:- GAM of Gold = 197 g

$\Rightarrow$ 197 g of Gold has = 6.022×10^{23} atoms.

1 g of Gold has = $\frac{6.022 \times 10^{23}}{197} \times 1$

$= 3.06 \times 10^{21}$ atoms.